

Literature Answer: Countable Compact Free Spaces Have MAP

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Status. `literature_already_answered`. The second closing question in arXiv:1201.0847 has a negative answer in later literature.

Source question. In Godefroy–Ozawa, *Free Banach spaces and the approximation properties*, arXiv:1201.0847, the authors close the paper with several questions. Question 2 asks:

Is there a countable compact space K such that $\mathfrak{F}(K)$ fails (AP)?

Later answer. Dalet, *Free Spaces over Countable Compact Metric Spaces*, arXiv:1307.0735, proves in Theorem 3.1 that if K is a countable compact metric space, then $\mathcal{F}(K)$ has the metric approximation property (MAP). Since MAP implies AP, there is no countable compact metric space K for which $\mathcal{F}(K)$ fails AP. The introduction of arXiv:1307.0735 explicitly states that this gives a negative answer to Question 2 in Godefroy–Ozawa.

Adjacent uniformly-discrete subcase. The first closing question in arXiv:1201.0847 is Kalton’s problem asking whether $\mathfrak{F}(M)$ has BAP for every uniformly discrete metric space M . This packet does not solve that problem. It records one nearby known positive subcase: Dalet, *Free spaces over some proper metric spaces*, arXiv:1404.3939, Theorem 2.6 proves MAP for countable proper metric spaces. In particular, every proper uniformly discrete metric space is locally finite and countable, so this covers the proper/locally finite uniformly-discrete case. The arbitrary uniformly-discrete problem remains open; for example, Cuth–Kalenda, arXiv:2604.01875, still describe it as a major open problem while proving that uniformly-discrete free spaces are 3-Schur.

Why this is not new. The decisive source arXiv:1307.0735 explicitly identifies the Godefroy–Ozawa question and states that its theorem answers it. The additional proper-metric paper is included only as duplicate-prevention context for the adjacent uniformly-discrete BAP question.

References.

- G. Godefroy and N. Ozawa, *Free Banach spaces and the approximation properties*, arXiv:1201.0847.
- A. Dalet, *Free Spaces over Countable Compact Metric Spaces*, arXiv:1307.0735.
- A. Dalet, *Free spaces over some proper metric spaces*, arXiv:1404.3939.
- M. Cuth and O. F. K. Kalenda, *Lipschitz-free spaces over uniformly discrete metric spaces are 3-Schur*, arXiv:2604.01875.